

Unit 2 — A compactified scale-duality field: closure of the static sector and falsifiable natural inflation

Locked non-minimal coupling and inter-sector signature $\beta(\text{CMB}) = \beta(\text{closure})$

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Abstract

Context. Singularity resolution and the origin of inflation are two requirements of any Planck-regime physics. **Problem.** We study whether a single scale-duality field ψ — arising from the compactification of the scale axis into a circle, coupled non-minimally to curvature via $f \propto 1 + \beta \cos \psi$ and carrying a bounded potential $V = \alpha(1 - \cos \psi)$ — can both regularise the Schwarzschild singularity and drive inflation. **Method.** The field is treated in two regimes: a static radial profile (black holes) and a homogeneous cosmological background (inflation), with passage to the Einstein frame. All tensor calculations are verified by symbolic computation and the equations integrated numerically. **Results.** (i) The unified pole generates an exact de Sitter core ($w = -1$, $K = 24H^4$ finite) whose radial stability is governed by an exact threshold $\beta_c = -1/3$. (ii) In horizon-penetrating coordinates, $\psi \equiv -\pi$ is exactly Schwarzschild–de Sitter, and the « rolling » branch develops a singular horizon: the static sector produces no regular black hole. (iii) The cosmological sector is a natural inflation with a topologically locked non-minimal coupling; it provides predictions (n_s, r) , and — α being fixed by the scalar amplitude A_s — an inflation scale $V^{1/4} \sim 10^{16}$ GeV and a running prediction $dn_s/d \ln k \approx -6.5 \times 10^{-4}$. (iv) The field being S^1 -valued ($\pi_1 = \mathbb{Z}$), the static profile is an unprotected half-soliton ($N = \frac{1}{2}$), whose winding topologically recovers the closure and identifies a scale soliton $N = 1$ of mass $8\sqrt{\omega\alpha}$. **Conclusion.** The framework is probably not a theory of regular black holes, but a predictive inflationary model falsifiable by LiteBIRD and the Simons Observatory. Its distinctive signature is no isolated cosmological observable — shared with [14, 16, 17] — but the inter-sector correlation: the same β must fit (n_s, r) and satisfy the black-hole closure thresholds.

Keywords: regular black holes; de Sitter core; scalar-tensor gravity; non-minimal coupling; natural inflation; Eddington–Finkelstein coordinates; topological scale charge; sine-Gordon soliton; cosmic microwave background.

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Epistemic status: [proof] exact symbolic computation · [numerical] reproducible integration · [conjecture] plausible, not proven · [heuristic] analogy · [speculation] open · [open] to be established · [to check] external ref. No conjecture is presented as a theorem.

Preliminary note. The present work proceeds from an initial intuition — the compactification of the scale axis into a circle — developed independently, without knowledge of the relevant cosmological literature. The bibliographic survey was undertaken only once the framework was established; it turns out that certain sectors overlap with published results, notably natural inflation with a periodic non-minimal coupling proportional to the potential [14], its scalar-tensor extension [16], and the $(n_s, r, \text{running})$ analyses against BICEP/Keck [17]. These overlaps, not anticipated, illustrate an independent convergence; we cite them explicitly so as to position honestly the specific contribution of this work, which lies in the unified framing of a single scale field (black-hole and cosmological regimes) and the topological reading of the static sector.

I Introduction

I.A Context and physical motivation

Two requirements weigh on any Planck-regime physics: removing the singularities of general relativity, and accounting for the primordial accelerated-expansion phase. Regular black-hole metrics [1, 2, 3] show that a central singularity can be replaced by a de Sitter core of finite curvature ($w = -1$), at the cost of a strong-energy-condition violation permitted by the singularity theorems [4, 5]. Symmetrically, inflation [6, 7] requires a scalar field slowly rolling in a flat potential. The structuring question is whether a single fundamental ingredient generates both phenomena, rather than postulating them separately.

The compactified scale-duality framework [15] proposes such an ingredient. Its central hypothesis is that the logarithmic scale axis $u = \ln(\lambda/\ell_P)$, usually non-compact, is in fact a circle: through the stereographic projection $\psi = 2 \arctan u \in (-\pi, \pi]$, the Planck scale ($\psi = 0$, pole P) and a « unified » scale ($\psi = \pm\pi$ identified, pole ∞^*) become two poles of one manifold. The scale field ψ couples non-minimally to curvature.

I.B State of the art and gap

Regular de Sitter cores are, in the literature, geometric ansätze: the $w = -1$ source is inserted by hand [1, 2, 3]. The no-hair theorems [8] strongly constrain static scalar fields around horizons. On the cosmology side, natural inflation [7] is under tension: Planck+BICEP/Keck [10, 11] push its tensor-to-scalar ratio r beyond the bounds for accessible decay constants, while non-minimal Higgs-like couplings [9] lower r . The gap filled is the study of a fundamental field whose non-minimal coupling is not free but locked by the same trigonometric structure as the potential, examined simultaneously in the black-hole and cosmological regimes.

I.C Contributions

We establish four results: (1) an exact de Sitter core and an exact stability threshold $\beta_c = -1/3$ (Theorem 1); (2) the closure of the static sector — $\psi \equiv -\pi$ is exactly Schwarzschild–de Sitter and the rolling branch is singular, hence no regular black hole (Result 1); (3) the identification of the cosmological sector with a natural inflation with locked coupling, with predictions (n_s, r) , α fixed and running, and a first constraint on β (Result 2); (4) a topological scale charge ($\pi_1(S^1) = \mathbb{Z}$) whose static sector is an unprotected half-soliton (Result 3).

II Formal framework

II.A Postulates and definitions

Postulate 1 (compactification). *The fundamental manifold is $\mathcal{M} = M_{3+1} \times S_\psi^1 \times S_\chi^1$, the scale axes being circles via $\psi = 2 \arctan[\ln(\lambda/\ell_P)]$. The poles $\psi = \pm\pi$ are identified (∞^*); $\psi = 0$ is the Planck point P .*

Postulate 2 (symmetry and regularity). *The action is invariant under the Klein group $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ acting by $(\psi, \chi) \mapsto (-\psi, -\chi)$, and all its functions are bounded on the circle (hence $\cos \psi$ rather than $\cosh u$).*

In reduced Planck units ($M_{\text{Pl}} = 1$, $\hbar = c = 1$, signature $-+++$) and freezing the temporal axis $\chi \equiv 0$:

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} f(\psi) R - \frac{1}{2} \omega (\partial\psi)^2 - V(\psi) \right], \quad f = 1 + \beta \cos \psi, \quad V = \alpha(1 - \cos \psi), \quad (1)$$

with $\omega = f_s^2 > 0$, $|\beta| < 1$, $\alpha > 0$. One has $V(0) = 0$ (minimum, P) and $V(-\pi) = 2\alpha$ (maximum, ∞^*).

II.B Mathematical construction

[proof] (verified by symbolic computation) Variation gives the scalar-tensor system

$$f G_{\mu\nu} = \omega \partial_\mu \psi \partial_\nu \psi - g_{\mu\nu} \left[\frac{1}{2} \omega (\partial \psi)^2 + V \right] + \nabla_\mu \nabla_\nu f - g_{\mu\nu} \square f, \quad \omega \square \psi = V'(\psi) - \frac{1}{2} f'(\psi) R. \quad (2)$$

In the Jordan frame, we use the static metric $ds^2 = -e^{2\Phi} dt^2 + e^{2\Lambda} dr^2 + r^2 d\Omega^2$ and the FLRW metric $ds^2 = -dt^2 + a^2 d\mathbf{x}^2$. In the Einstein frame ($\tilde{g} = fg$), the action becomes that of a canonical field, with

$$U(\psi) = \frac{V}{f^2}, \quad K(\psi) = \frac{\omega}{f} + \frac{3}{2} \left(\frac{f'}{f} \right)^2, \quad d\varphi = \sqrt{K} d\psi. \quad (3)$$

II.C Structural properties

[proof] (established in [15]) (i) The action is finite everywhere, ∞^* included. (ii) The $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry forbids $\psi\chi$ mixing: diagonal mass matrix, no Goldstone mode (impossible for a discrete symmetry). (iii) At the symmetric vacuum $P, V = 0$ so $\Lambda = 0$. (iv) At frozen field, Einstein gravity emerges with $G_{\text{eff}} = 1/(8\pi f)$, bounded between $G/(1+\beta)$ (P) and $G/(1-\beta)$ (∞^*): the natural sign of the coupling is $\beta \geq 0$.

(v) [proof] (B1) The Einstein-frame field-space metric $K(\psi)$ is positive definite $\Leftrightarrow f > 0 \Leftrightarrow |\beta| < 1$ (mono-axis): it is the sum of a kinetic part (ω/f) positive definite and a non-minimal part $\frac{3}{2}(f'/f)^2$ positive semi-definite (rank 1). The non-minimal coupling can therefore only add positivity — it never creates a ghost. As $f \rightarrow 0$ ($|\beta| \rightarrow 1$) the Einstein frame degenerates; the band $|\beta| < 1$ is open.

III Main results

III.A Exact de Sitter core and stability threshold $\beta_c = -1/3$

On the static metric, R is eliminated by the trace of the tensor equation; [proof] the three residuals (tt, rr, scalar) are then exactly linear in $(\Lambda', \Phi', \psi'')$ (Appendix A).

Theorem 1 (de Sitter core and coupling threshold). *At the pole ∞^* ($\psi = -\pi, \psi' = 0$), the source reduces to a pure vacuum energy and the spacetime is locally de Sitter:*

$$H^2 = \frac{2\alpha}{3(1-\beta)}, \quad K = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} = 24H^4 \text{ (finite)}, \quad \rho = 2\alpha, \quad p_r = -2\alpha, \quad w = -1. \quad (4)$$

The radial fluctuation $\psi = -\pi + \delta$ obeys $\omega \square \delta = \kappa^2 \omega \delta$ with $\kappa^2 = \alpha(1+3\beta)/[\omega(\beta-1)]$. Since $|\beta| < 1$, ∞^* is radially stable (trapped) for $\beta > -1/3$ and unstable (escaping) for $\beta < -1/3$, with the exact threshold $\boxed{\beta_c = -1/3}$.

Note (reduction). This threshold $\beta_c = -1/3$ is specific to the *static mono-axis reduction* (χ frozen). A two-field cosmological treatment along the diagonal $\psi = \chi$ shifts the threshold to $-1/6$; the closure conclusion (no static regular black hole) is robust regardless of the reduction.

Consequence (Derrick obstruction). For the natural sign $\beta \geq -1/3$, the field cannot connect ∞^* to the vacuum: the gradient energy being positive, the maximum of V becomes a minimum of the effective mechanical potential $-V$, and the field remains trapped there. The escape window $\beta < -1/3$ exists only through the non-minimal term $\frac{1}{2} f' R$.

III.B Closure of the static sector in Eddington–Finkelstein coordinates

The static chart ends at a horizon ($e^{-2\Lambda} \rightarrow 0$); [numerical] the outgoing integration from the core shows, in both regimes, that a horizon forms exactly at the de Sitter radius $r_H = 1/H = \sqrt{3(1-\beta)/(2\alpha)}$ (numerical/theory ratio = 1.0000). To cross it, one passes to the ingoing chart $ds^2 = -A dv^2 + 2B dv dr + r^2 d\Omega^2$, regular as long as B stays finite (Appendix A).

Result 1 (no regular black hole). *[proof]* $\psi \equiv -\pi$ is an exact solution: $A(r) = 1 - 2M/r - H^2 r^2$, $B \equiv 1$ (Schwarzschild-de Sitter). The trapped regime $\beta > -1/3$ is therefore asymptotically de Sitter, not an asymptotically flat black hole. *[numerical]* In the escaping regime $\beta < -1/3$, the field rolls from $-\pi$ towards 0, but $B = e^{\Phi+\Lambda}$ collapses to 0 at the putative de Sitter radius: the Misner–Sharp mass $m = (r/2)(1 - A/B^2)$ becomes negative and the Ricci scalar diverges at finite radius. No spherical static solution interpolates a regular de Sitter core and an asymptotically flat exterior.

r	ψ	m (Misner–Sharp)	$g_{rr} = A/B^2$	$ R $ (Ricci)
0.01	-3.132	$+5 \times 10^{-10}$	1.00	10^{-2}
47.1	-1.4	-3×10^7	1.3×10^6	6×10^{11}
75.5	-0.002	-10^{56}	3×10^{54}	10^{73}

Table 1: Escaping regime ($\beta = -0.7$, $\alpha = 10^{-4}$): the rolling branch is singular.

III.C The cosmological sector: natural inflation with locked coupling

In homogeneous cosmology the gradient becomes temporal: $(\partial\psi)^2 = -\dot{\psi}^2$. *[proof]* The potential curvature at ∞^* is $V''(-\pi) = -\alpha < 0$: ∞^* is an unstable summit, hence the field rolls down towards P (Fig. 1).

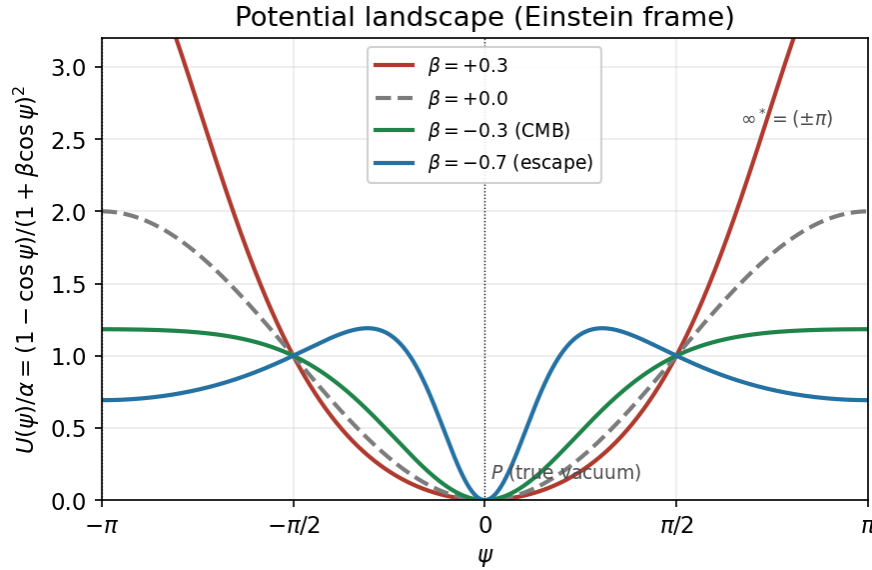


Figure 1: *[numerical]* Einstein-frame potential landscape, $U(\psi)/\alpha = (1 - \cos \psi)/(1 + \beta \cos \psi)^2$ (exact closed form). The true vacuum P is at $\psi = 0$ ($U = 0$), the unified pole ∞^* at $\psi = \pm\pi$. For the CMB-favoured value ($\beta = -0.3$), the inflaton rolls from ∞^* down to P along a smooth hill; the locked coupling $f = 1 + \beta \cos \psi$ shapes it.

Result 2 (natural inflation, predictions, α fixed and constraint on β). *With the canonical field $\varphi = f_s \psi$, the potential reads $V = \alpha[1 - \cos(\varphi/f_s)]$: this is natural inflation [7] with constant f_s , endowed with a locked non-minimal coupling $f = 1 + \beta \cos \psi$ (same β as gravity); this periodic structure was studied independently by [14, 16]. *[numerical]* The FLRW integration starts at ∞^* (de Sitter phase, $H^2 = 2\alpha/[3(1 - \beta)]$, identical to the core scale of Result 1) and rolls to P ($\Lambda = 0$), with a graceful exit. The slow-roll predictions in the Einstein frame (Table 2) give: for $\beta \geq 0$, $r \gtrsim 0.08$ (excluded); for $\beta \approx -0.3$, $r \approx 0.02$ – 0.05 and $n_s \approx 0.96$ (borderline). CMB data constitute the first constraint on β in this framework.*

β	f_s (M_{Pl})	n_s	r	status
0 (natural)	7	0.960	0.078	r too large
+0.3	7	0.949	0.128	excluded
+0.6	7	0.930	0.193	excluded
-0.3	5	0.958	0.017	compatible
-0.3	7	0.960	0.036	borderline

Table 2: Predictions (n_s, r) at $N = 55$ e-folds (target Planck+BK: $n_s = 0.9649 \pm 0.004$, $r < 0.036$).

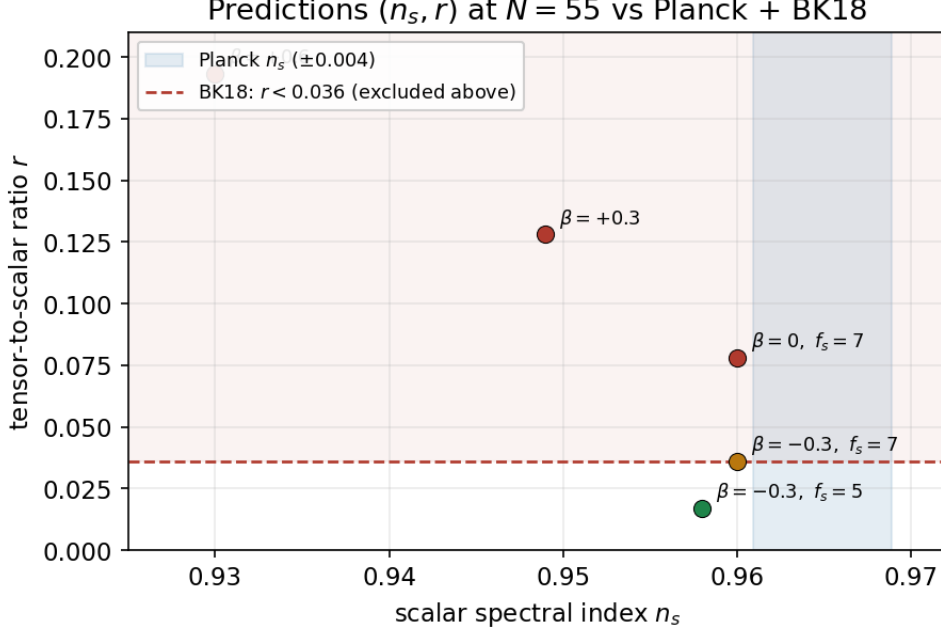


Figure 2: [numerical] Predictions (n_s, r) at $N = 55$ (points from Table 2) against Planck ($n_s = 0.9649 \pm 0.004$) and BK18 ($r < 0.036$, excluded above). The sign of β shifts r : $\beta \geq 0$ is excluded; the viable window is $\beta \approx -0.3$ ($r \approx 0.02$ – 0.04 , $n_s \approx 0.96$).

[numerical] **(A1) α fixed and running.** Second-order slow-roll with the field-space metric $K(\psi)$ reproduces Table 2 to better than 0.5%, and fixes α via the scalar amplitude $A_s = 2.1 \times 10^{-9}$: $\alpha \approx 0.5$ – $1.2 \times 10^{-9} M_{\text{Pl}}^4$ (central case $\beta = -0.3$, $f_s = 7$: $\alpha = 1.18 \times 10^{-9}$), i.e. an inflation scale $V^{1/4} \sim 1$ – 2×10^{16} GeV (GUT). A prediction stable over the band: the running $dn_s/d \ln k \approx -6.5 \times 10^{-4}$ — compatible with Planck (-0.004 ± 0.007), below the current threshold, possibly accessible to the next generation, but not distinctive: it belongs to the class [14, 16, 17]. Remainder [open]: a mode-by-mode Mukhanov–Sasaki resolution, expected at the sub-percent level.

Status of the $\Delta \sim 0.044$ signature. [numerical] The background trajectory is monotonic (half-period of the cosine): the primordial spectrum is smooth, without log-periodic feature. The $\Delta \sim 0.044$ signature evoked in [15] is not reproduced by the minimal dynamics; it would require an additional mechanism (a wound trajectory $N \geq 1$, or an unfrozen χ axis). [open]

III.D Topological scale charge and half-soliton

The field ψ being S^1 -valued, its configurations are classified by $\pi_1(S^1) = \mathbb{Z}$. [proof] In the effective 1+1 D reduction, the current $j^\mu = (1/2\pi)\varepsilon^{\mu\nu}\partial_\nu\psi$, of charge $N = (1/2\pi)(\psi_f - \psi_i)$, is identically conserved ($\partial_\mu j^\mu \equiv 0$), independently of the equations of motion, the metric, f and V . On the circle, $V = 2\alpha \sin^2(\psi/2)$ is a sine-Gordon with a unique vacuum. The charge is integer iff both endpoints rest on the vacuum P .

Result 3 (unprotected half-soliton). *[proof] The profile of Result 1, connecting the core $\psi(0) = -\pi$ (∞^*) to the asymptotic $\psi(\infty) = 0$ (P), carries charge $N = \frac{1}{2}$. With one endpoint lying on the maximum of V , it is not protected: the arc $\infty^* \rightarrow P$ is contractible, and its contraction coincides with the « rolling ». The closure of the static sector follows without integration. The unique non-trivial integer charge ($\psi : 0 \rightarrow 2\pi$) is the soliton $N = 1$, of Bogomol’nyi mass $M_{N=1} = 8\sqrt{\omega\alpha}$ (Appendix D).*

Observational link. [conjecture] In cosmology, N counts the periods travelled by the inflaton. The monotonic trajectory ($N = \frac{1}{2}$) gives a smooth spectrum; a wound trajectory ($N \geq 1$, crossing ∞^*) would be the only source of log-periodic oscillations. The test is a Mukhanov–Sasaki resolution on an $N \geq 1$ background.

IV Physical implications

IV.A Interpretation

The same scale field is benign in time and obstructed in space. In cosmology, ∞^* is a hill from which the field rolls: it generates inflation and a natural exit towards $\Lambda = 0$. In the static case, ∞^* is a well: the field stays trapped and the spacetime is merely Schwarzschild–de Sitter. The remarkable fact is that the same de Sitter scale $H^2 = 2\alpha/[3(1 - \beta)]$ appears as the cosmological horizon of the static core and as the inflationary Hubble rate: a signature of the common origin of the two phenomena.

IV.B Limitations and epistemic honesty

(i) The closure (Result 1) concerns only spherical static configurations; a dynamical collapse could produce a transient regular core (treated in a companion work). (ii) The predictions (Result 2) rest on slow-roll; α is fixed by A_s on this validated basis (running -6.5×10^{-4}), a mode-by-mode Mukhanov–Sasaki treatment remaining a sub-percent refinement (not a prerequisite). (iii) The $\Delta \sim 0.044$ signature is not established. (iv) The constants α, β, f_s are not derived from first principles; β is only constrained by the data. None of these points is presented as resolved. [open]

V Experimental protocols

V.A Main discriminating test: the tensor-to-scalar ratio

The model predicts, in its viable window ($\beta \approx -0.3$, $f_s \approx 5\text{--}7 M_{\text{Pl}}$, $N = 50\text{--}60$), a ratio $r \approx 0.02\text{--}0.05$ correlated with $n_s \approx 0.96$. The measurement of r by LiteBIRD [12] and the Simons Observatory [18], with sensitivity $\sigma(r) \sim 10^{-3}$, is the discriminating test¹: a non-detection at $r < 0.01$ would exclude the model; a detection at $r \sim 0.03$ with $n_s \approx 0.96$ would corroborate it.

V.B The distinctive signature: inter-sector consistency

[the test that distinguishes this framework] No isolated cosmological observable (n_s , r , running) distinguishes this model from non-minimal natural inflation [14, 16, 17]. Its distinctive signature is a correlation: the sign and value of β favoured by the CMB ($\beta \approx -0.3$) must be confronted with the black-hole sector, where the threshold $\beta_c = -1/3$ governs the stability of the de Sitter core. A single value of β must fit (n_s, r) and lie on the right side of the closure threshold. A persistent

¹The CMB-S4 project (DOE–NSF), long considered the reference instrument for this test, has been halted: on 9 June 2025 the DOE and NSF declared they could no longer support the project (orderly shutdown), which is therefore no longer funded [13]. The falsifiability of the prediction remains, carried by the Simons Observatory (Chile, operating), LiteBIRD (launch targeted ~ 2032 , designed for $\sigma(r) < 10^{-3}$) and BICEP/Keck (ongoing). [to check]

tension between the two sectors would be a structural failure; their agreement would be the true corroboration of the unified framework.

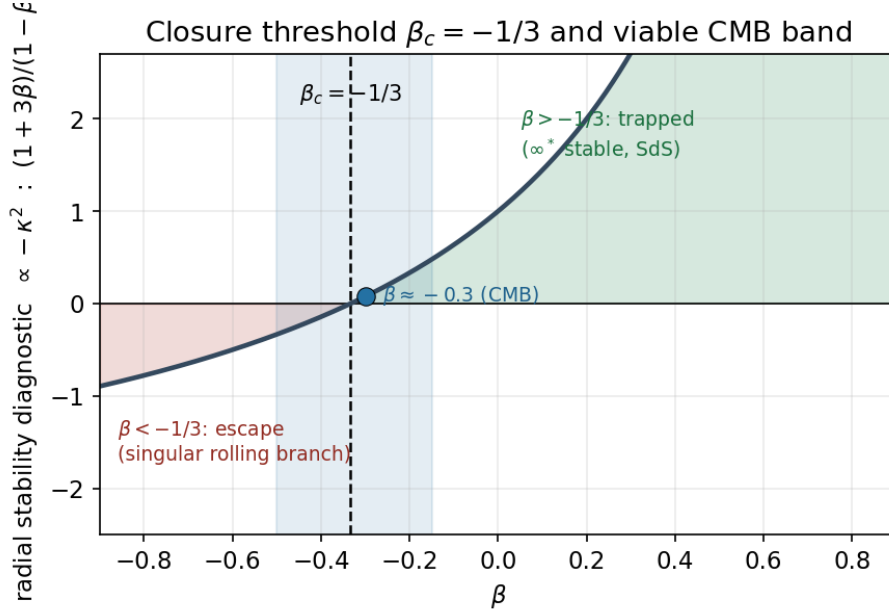


Figure 3: [numerical] Radial-stability diagnostic at the pole ∞^* ($\propto -\kappa^2$, closed form $(1+3\beta)/(1-\beta)$), changing sign at the exact threshold $\beta_c = -1/3$: $\beta > -1/3$ trapped (stable de Sitter core, Schwarzschild–de Sitter), $\beta < -1/3$ escaping (singular rolling branch). The CMB-favoured value ($\beta \approx -0.3$, band -0.5 to -0.15) falls *just* on the trapped side: this is the visualisation of the signature $\beta(\text{CMB}) = \beta(\text{closure})$.

V.C Falsifiability and bounds

The model lives in a narrow band of the (n_s, r) plane. Current bounds: $\beta \geq 0$ already excluded ($r \gtrsim 0.08 > 0.036$); $\beta \lesssim -0.6$ excluded (insufficient inflation). Only the band $-0.5 \lesssim \beta \lesssim -0.15$ survives, fully testable by the 2030 horizon.

VI Discussion and comparison

VI.A Relation to competing approaches

Unlike phenomenological regular black holes [1, 2, 3], where the $w = -1$ source is postulated, we derive the de Sitter core from an action and show that it does not connect to a flat exterior in the static case — in agreement with the spirit of the no-hair theorems [8]. For inflation, the locked non-minimal coupling distinguishes the model from bare natural inflation [7] and from Higgs-like inflation [9]: the coupling is not free but fixed by the same structure as the potential, turning a tuning parameter into a quantity constrained by gravity. The periodic structure itself joins [14, 16], from which we differ by the inter-sector framing.

VI.B Open questions

Four directions, outside the scope of the present work [open]: (i) the correlation $\beta(\text{CMB}) = \beta(\text{closure})$ — posed here as a signature (§ V.B) — deserves a joint quantitative study; (ii) the mode-by-mode Mukhanov–Sasaki resolution to confirm $(n_s, r, \text{running})$ at the sub-percent level; (iii) the possible origin of log-periodic oscillations via a multi-period trajectory ($N \geq 1$) or an unfrozen temporal axis χ ; (iv) the dynamical sector of gravitational collapse, the only remaining avenue for a black-hole regularisation (companion work).

VII Conclusion

A single scale-duality field, treated in two regimes, produces two opposite and both instructive verdicts. In the static case, it fails to regularise the singularity: the unified pole generates an exact de Sitter core but one inevitably frozen (Schwarzschild–de Sitter) or singular, and no regular black hole exists — with an exact threshold $\beta_c = -1/3$ as a by-product. In cosmology, the same field generates a natural inflation with locked coupling, predictive and falsifiable, whose α is fixed (GUT scale) and which predicts a running $dn_s/d\ln k \approx -6.5 \times 10^{-4}$. The distinctive signature of the framework is not an isolated cosmological observable, but the correlation $\beta(\text{CMB}) = \beta(\text{closure})$. The programme reorients from a theory of black holes towards a theory of cosmological history, testable by observation for the first time.

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A Technical details

Static system. With R eliminated by the trace, the (tt) , (rr) and scalar residuals are linear in $(\Lambda', \Phi', \psi'')$ and solved pointwise by 3×3 inversion; the mixed Einstein tensor is validated against the Schwarzschild vacuum. **Eddington–Finkelstein chart.** For $ds^2 = -A dv^2 + 2B dv dr + r^2 d\Omega^2$, one has $g_{vv} = 0$, $g_{vr} = 1/B$, $g_{rr} = A/B^2$, $\sqrt{-g} = Br^2$, and $G_{rr} = 2B'/(rB)$. The 3×3 system for (A', B', ψ'') is linear (linearity residual exactly zero). The horizon is a characteristic surface ($g_{rr} = A/B^2 \rightarrow 0$); regularity imposes $\Pi(r_H) = 0$ with $\Pi = (Ar^2/B)\psi'$. **Cosmology.** The FLRW background solves the Friedmann constraint $3fH^2 = \frac{1}{2}\omega\dot{\psi}^2 + V - 3H\dot{f}$ and the acceleration equation.

B Dimensional checks

In units $\hbar = c = 1$, $M_{\text{Pl}} = 1$: $[\alpha] = [V] = \ell_P^{-4}$, $[H^2] = \ell_P^{-2}$, consistent with $H^2 = 2\alpha/[3(1 - \beta)]$ since f carries the dimension M_{Pl}^2 ($f(-\pi) = (1 - \beta)M_{\text{Pl}}^2$), whence $[H^2] = [\alpha]/M_{\text{Pl}}^2 = \ell_P^{-2}$. β and ψ are dimensionless; f_s has the dimension of M_{Pl} . $K = 24H^4$ has dimension ℓ_P^{-4} , as expected for $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$.

C Quantitative comparison with the literature

For bare natural inflation ($\beta = 0$), our values ($n_s = 0.960$, $r = 0.078$ at $f_s = 7 M_{\text{Pl}}$, $N = 55$) reproduce the standard results [7, 10] and confirm the known tension with [11]. The effect of the locked coupling ($\beta < 0$ lowers r ; $\beta > 0$ raises it) is analogous in direction to that of a non-minimal Higgs-like coupling [9], but of a magnitude controlled by the single gravitational parameter β . The predicted running ($\approx -6.5 \times 10^{-4}$) is consistent with the $(n_s, r, \text{running})$ analyses of this class [17]. The de Sitter core ($w = -1$, $K = 24H^4$) coincides with the interior structure of regular black holes [2, 3], but without the asymptotically flat matching those models postulate.

D Topological current and Bogomol’nyi bound

For $\psi \in S^1$, $j^\mu = (1/2\pi)\varepsilon^{\mu\nu}\partial_\nu\psi$ satisfies $\partial_\mu j^\mu \equiv 0$ by antisymmetry, without use of the equations of motion; $N = (1/2\pi)\oint d\psi$ is a homotopy invariant. The flat static energy $E = \int [\frac{1}{2}\omega\psi'^2 + V]dr$ admits the bound $E \geq \int \sqrt{2\omega V}|d\psi|$, saturated for $\sqrt{\omega}\psi' = \sqrt{2V}$, i.e. the kink $\psi(r) =$

$4 \arctan e^{\sqrt{\alpha/\omega}(r-r_0)}$, of width $\sim \sqrt{\omega/\alpha}$, and $M_{N=1} = 2\sqrt{\omega\alpha} \int_0^{2\pi} |\sin(\psi/2)| d\psi = 8\sqrt{\omega\alpha}$. The half-soliton ($\psi : -\pi \rightarrow 0$) has energy $4\sqrt{\omega\alpha}$ but connects the maximum to the vacuum: it is not BPS and contracts (§ III.D).

Reproducibility

The numerical results and the symbolic verification are reproduced by two standalone scripts (NumPy/SymPy/SciPy/Matplotlib), with sections coded so as to be specific to this document:

- [G1] Einstein-frame geometry ($U = V/f^2$, $K = \omega/f + \frac{3}{2}(f'/f)^2$) and positivity $K > 0 \Leftrightarrow f > 0 \Leftrightarrow |\beta| < 1$ (mono-axis).
- [G2] closure threshold: sign of $\kappa^2 = \alpha(1 + 3\beta)/[\omega(\beta - 1)]$, exact root $\beta_c = -1/3$.
- [G3] de Sitter core: $H^2 = 2\alpha/[3(1 - \beta)]$, $r_H = 1/H$, Kretschmann $= 24H^4$.
- [G4] inflation: (n_s, r) at $N = 55$ (slow-roll, metric K) — reproduces Table 2; α fixed by A_s ($V^{1/4} \approx 1.4 \times 10^{16}$ GeV), running.
- [G5] topology: charge N , soliton $M_{N=1} = 8\sqrt{\omega\alpha}$, half-soliton $4\sqrt{\omega\alpha}$.

Files: `U2_GRAV_reproducible.py` (calculations) and `Gravite_figures.py` (Fig. 1–3). *Convention*: α cancels in the (n_s, r) ratios and fixes only the absolute scale; $M_{\text{Pl}} = 1$ unless otherwise stated. The prefix `U2_GRAV_` and the `[G-]` codes are specific to this document.

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